

ASGARD 2017 – Survey vs Processing

Comparison report · 2026-07-02

A living summary of differences between the survey (0.2 um, 11 clusters, 181 samples) and processing (0.2/3/20 um, 4 clusters, 78 samples) datasets.

Contents:

1. Environmental variables (6-variable boxplot comparison)
2. db-RDA (5-variable constrained ordination)

Note: processing environmental values are triplicated across filter fractions; they are de-duplicated to unique station x depth for fair environmental comparison.

1. Environmental variables (6-var)

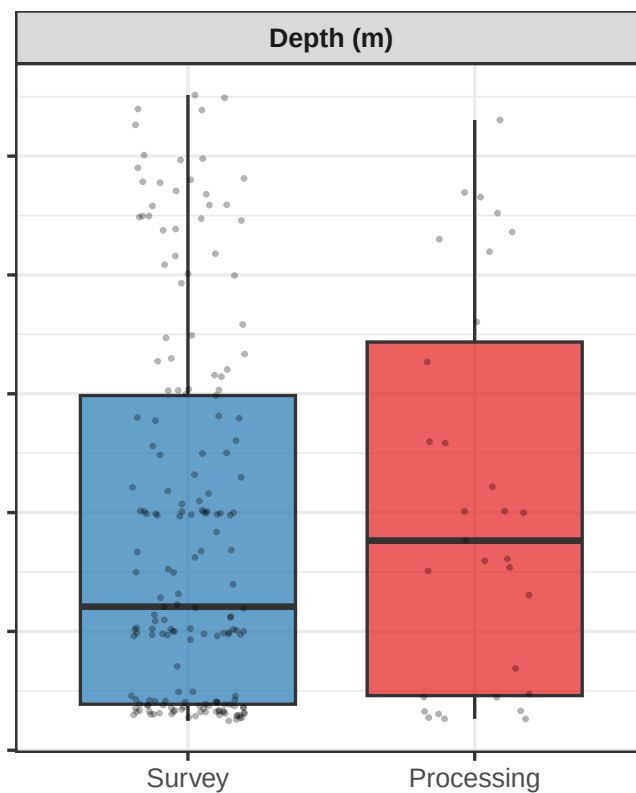
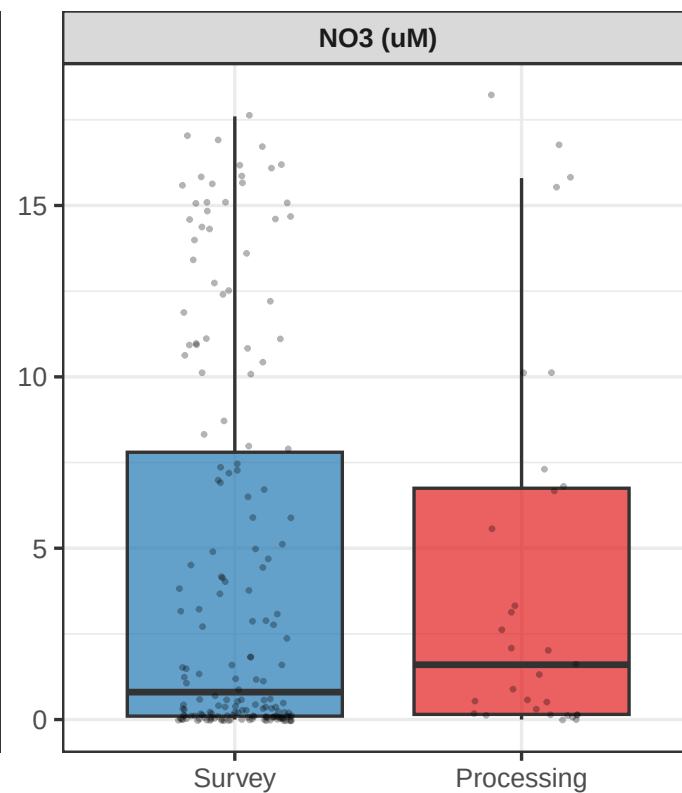
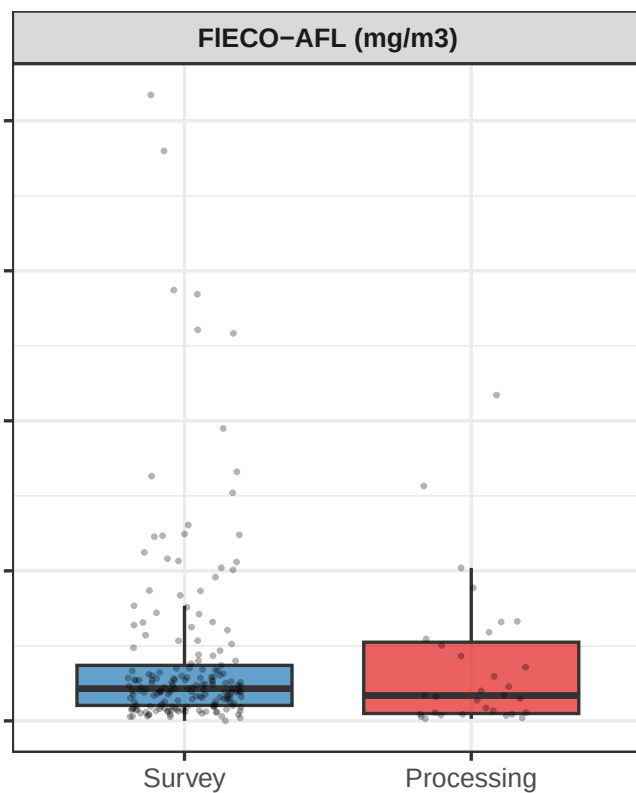
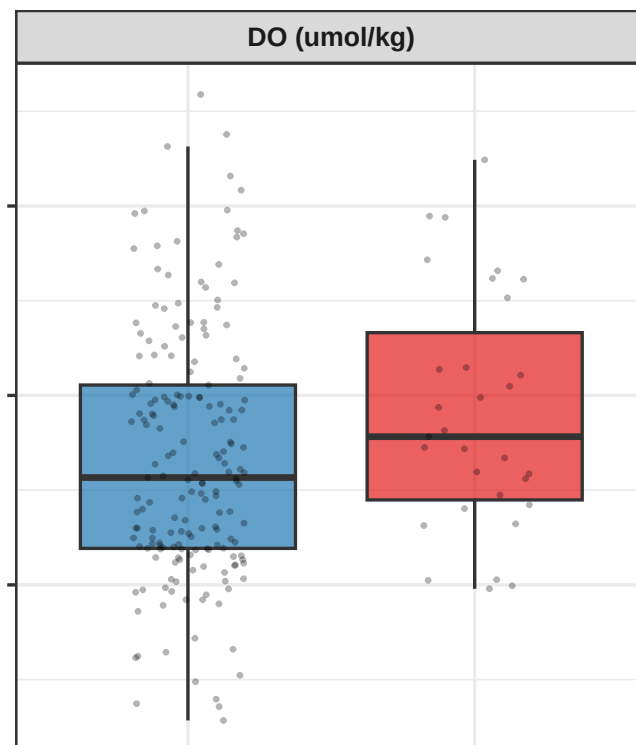
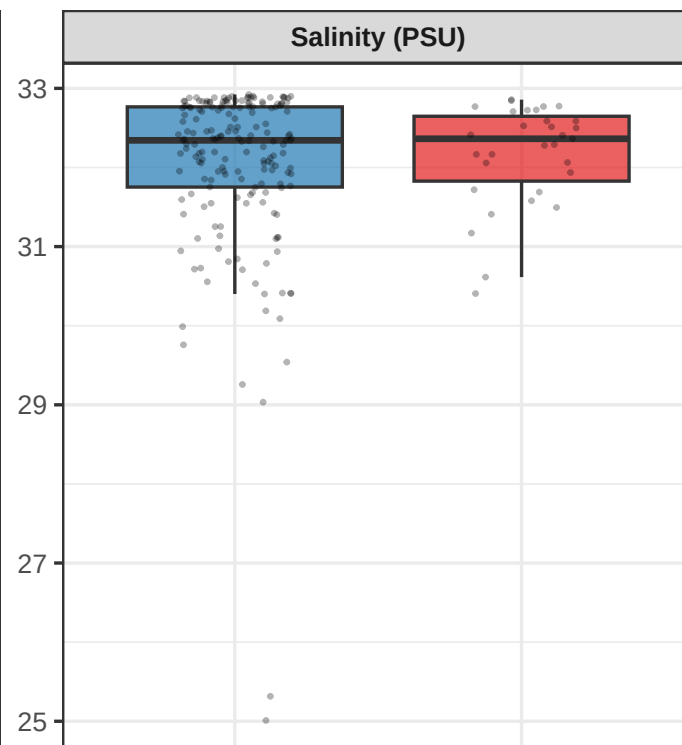
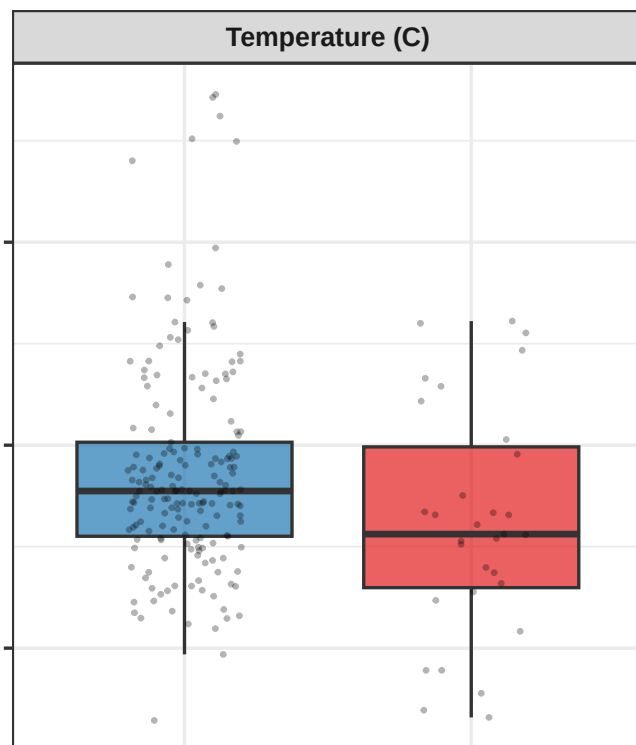
Q: Do survey and processing sample different environmental conditions?

Method: Wilcoxon rank-sum per variable; processing de-duplicated to unique water samples (78 → 32); Bonferroni $\alpha = 0.05/6 = 0.0083$.

Result: NO variable differs significantly after correction (all ns).

- Environments broadly OVERLAP between the two datasets.
- Survey spans a WIDER range of salinity and temperature (more diverse water masses).
- Temperature is closest to differing (survey slightly warmer).

6 environmental variables: Survey vs Processing



1. Environmental variables – Wilcoxon (Bonferroni $\alpha=0.0083$)

Variable	Med_Survey	Med_Proc	Survey_range	Proc_range	p	sig
Temperature (C)	3.10	2.25	−1.4–10.9	−1.4–6.4	0.0194	ns
Salinity (PSU)	32.34	32.36	25.0–32.9	30.4–32.9	0.8450	ns
DO (umol/kg)	328.33	339.11	264.2–429.4	299.0–412.2	0.0590	ns
FIECO–AFL (mg/m3)	1.07	0.84	0.0–20.9	0.1–10.9	0.2330	ns
NO3 (uM)	0.80	1.60	0.0–17.6	0.0–18.2	0.4990	ns
Depth (m)	12.08	17.64	2.5–55.1	2.6–53.0	0.4340	ns

2. db-RDA (5-var constrained ordination)

Q: Which environmental variables structure the microbial community, and how much?

Model: capscale(Bray-Curtis ~ temp + salinity + DO + NO3 + FIECO), 4th-root ASV props.

Result:

- Both overall models are significant ($p = 0.001$); environment structures both.
 - SURVEY: all 5 variables significant (esp. salinity & NO3); adj R2 ~ 0.15.
 - PROCESSING: ONLY salinity significant; adj R2 ~ 0.10.
- => Survey community tracks multiple gradients; processing is driven mainly by salinity.

Caveats: marginal ANOVA is collinearity-sensitive; processing n is smaller (75 vs 174);

CAP axis % are defined differently in S05 vs P06 and are not directly comparable.

2. db-RDA – overall model

Dataset	n	ASVs	Constrained_pct	Adj_R2	Overall_F	Overall_p
Survey	174	258	20.2	0.148	7.00	0.001
Processing	75	221	17.5	0.101	2.66	0.001

2. db-RDA – per-variable marginal ANOVA (F, p)

Variable	Survey_F	Survey_p	Proc_F	Proc_p	Survey_sig	Proc_sig
temp	4.81	0.001	1.32	0.199	*	ns
salinity	8.44	0.001	3.15	0.005	*	*
DO	2.72	0.003	0.72	0.712	*	ns
NO3	8.20	0.001	1.32	0.193	*	ns
FIECO	2.33	0.012	1.66	0.102	*	ns

3. Stations (survey vs processing)

Survey: 62 stations, 181 samples (1–7 samples/station; 0.2 um filter only).

Processing: 12 stations, 78 samples (1–9 samples/station; station x depth x up to 3 filter fractions).

Shared station names (6): CBE1, CBW5, CL1, CL3, DBO3.3, IL2

→ sampled in BOTH datasets; usable for direct same-site comparison.

Tables: (3a) processing per-station counts; (3b) survey per-station counts (3-column layout).

3a. Processing – samples per station (12 stations)

Station	n_samples	n_depths	Shared
CBE1	9	3	yes
IL2	9	3	yes
CL3	8	2	yes
DBO2.4	8	3	
DBO3.3	8	3	yes
CBE9	7	2	
CL1	7	3	yes
IL1.1	7	3	
CBW5	5	3	yes
DBO3.8A	5	2	
CNL3	4	2	
incubation	1	1	

3b. Survey – samples per station (62 stations)

Station.1	n.1	Station.2	n.2	Station.3	n.3
DBO3.6	7	CNL1	3	SLY9	3
DBO3.4	5	CNL10	3	SLY9.5	3
CBW3	4	CNL12	3	CB1	2
CL1.5	4	CNL5	3	CBE10	2
CL2.5	4	CPL6	3	CBW12	2
CL3	4	CPL7	3	CBW14	2
CNL2	4	CPL8	3	CBW15	2
CNL9	4	DBO2.2	3	CL2	2
DBO1.10	4	DBO2.3	3	CNL7	2
DBO3.7	4	DBO2.36	3	CPL1	2
DBO3.8	4	DBO3.1	3	CPL2	2
KL2	4	DBO3.5	3	CPL3	2
KL4	4	IL1	3	DBO2.5	2
CBE1	3	IL2	3	DBO3.2	2
CBE2	3	IL7	3	DBO3.3	2
CBE3	3	SLY2	3	IL3	2
CBE5	3	SLY4	3	CBE11	1
CBW5	3	SLY5	3	CPL5	1
CBW9	3	SLY6	3	IL5	1
CL0	3	SLY7	3	IL9	1
CL1	3	SLY8	3		

4. ASV counts (survey vs processing)

Both datasets are drawn from the same 16S prokaryote ASV universe (3,076 ASVs).

Observed (≥ 1 read): Survey 442, Processing 378.

- Survey observes more, but it also has more samples (181 vs 81) $\rightarrow$ partly sampling effort.
- Processing additionally includes 3 and 20 μm (particle-associated) fractions.

Retained for clustering (max RA > 0.001 AND present in > 2 samples):

Survey 258, Processing 221; 178 shared ($\sim 59\%$ of the 301-ASV union).

Survey has more unique retained ASVs (80) than processing (43).

4a. ASV counts by level

Level	Survey	Processing
Full 16S pool (universe)	3076	3076
Observed in dataset (count>0)	442	378
Retained after filtering (clustering)	258	221

4b. Retained-ASV overlap (258 vs 221)

Set	ASVs
Shared	178
Survey-only	80
Processing-only	43
Union	301

5. Dataset summary – samples & ASVs

Consolidated overview of sample and ASV counts for the two datasets.

ASV filtering (identical criteria for both): read counts converted to within-sample relative abundance; ASVs retained if max relative abundance > 0.001 (0.1%) AND present in > 2 samples.

Datasets differ only in read threshold (survey ≥ 1000 , processing ≥ 5000) and filter fractions sampled (survey 0.2 μm only; processing 0.2/3/20 μm).

5. Samples & ASVs – survey vs processing

Metric	Survey	Processing
Samples (analysis set)	181	78
Filter fractions	0.2 um	0.2/3/20 um
Min 16S reads	1000	5000
Stations	62	12
16S ASV pool	3076	3076
Observed ASVs (≥ 1 read)	442	378
Retained ASVs (clustering)	258	221

6. Clustering procedure (survey & processing)

Common: community structure was summarised by hierarchical clustering. Bray–Curtis dissimilarities among samples were computed from the fourth–root–transformed ASV relative–abundance matrices (survey 181 x 258; processing 78 x 221), and samples were clustered using Ward's method (ward.D).

Survey: the dendrogram was cut at $k = 10$ and organised into a three–level hierarchy with three top–level divisions (A, B, C). Cluster C1b was further subdivided by re–clustering the division–C samples (Ward, $k = 6$), yielding 11 final clusters (A1, A2, B1, B2a, B2b, C1a, C1b1, C1b2, C2a, C2b1, C2b2).

Processing: the dendrogram was cut at $k = 4$; the four clusters were grouped into two divisions reflecting size–fraction behaviour (free–living = cluster 1; particle–associated = clusters 2–4).

Environmental validation (processing only): a top–down tree walk compared, at each node, the two child groups (each ≥ 5 samples) on the principal components of the scaled environmental variables (PCs explaining $\geq 90\%$ of variance) using Wilcoxon rank–sum tests with Benjamini–Hochberg correction; a node was split only if at least one PC differed significantly (adjusted $p < 0.05$). This yielded a single cluster (no environmentally significant split at the root).

7. Statistical tests – where each was used

Comprehensive inventory of statistical tests across the survey (S*) and processing (P*) scripts.

Notes:

- Tukey's HSD is used ONLY as the post-hoc for PERMDISP (betadisper).
- BH (Benjamini–Hochberg) is the multiple-comparison correction; the depth Kruskal–Wallis tests (S06, P04) instead use Bonferroni.
- PERMANOVA, db-RDA and Mantel use 999 permutations; PERMDISP is NOT permutation-based (overall = ANOVA on betadisper; pairwise = Tukey's HSD).
- VIF is a collinearity diagnostic, not a significance test.

7. Statistical tests inventory

Test	Scripts	Applied.to
Kruskal–Wallis	S03, S06, S07, P04	reads / alpha-div / richness / component-RA vs cluster or depth
Wilcoxon rank-sum	P03, S07, S18, S20, S25	tree-walk env-PC; pairwise alpha-div; cluster-specific ASV; strat.; transect
t-test (Welch/Student)	S06, S11, S20, S25	group means (e.g. delta-sigma ~ sea; transect shares)
Fisher exact	S06, S11, S20, P14	contingency tables (depth/region x cluster/station)
Chi-squared	S06, S11, P14	contingency tables (simulated p-values)
Tukey HSD	S10, S17, P09	post-hoc for PERMDISP (within-group dispersion)
PERMANOVA (adonis2)	S10, S17, P09	community composition ~ cluster / environment
PERMDISP (betadisper)	S10, S17, P09	homogeneity of multivariate dispersion
Mantel test	S10	community distance ~ environmental distance
db-RDA perm. ANOVA	S05, P06	constrained ordination significance (overall/margin/axis)
Indicator species (IndVal)	S12, S13	cluster indicator ASVs (16S = S12, 18S = S13)
BH / FDR correction	P03, S06, S07, S10, S17, S18, P09	multiple-comparison correction (pairwise / tree-walk tests)
VIF (diagnostic)	S05	collinearity check for db-RDA variable selection

8. Cluster x depth association (unified test)

Q: Is cluster membership associated with sampling depth (surf/mid/bottom)?

Both variables are categorical, so a contingency-table test is used (not Kruskal-Wallis).

Unified method applied identically to both datasets: Fisher's exact test with Monte-Carlo simulation ($B = 9,999$); chi-squared (Monte-Carlo) reported alongside for reference.

This harmonises the survey (S06) and processing (P14) tests, which previously differed slightly (survey used Monte-Carlo Fisher + per-cluster BH; processing used standard Fisher).

8. Cluster x depth – Fisher's exact (Monte-Carlo, B=9999)

Dataset	n	Clusters	Fisher_p	ChiSq_p	Significant
Survey	181	11	0.943	0.938	no
Processing	77	4	0.448	0.464	no

9. Environmental testing of cluster splits – why survey and processing d

Both pipelines DEFINE primary clusters identically (Bray–Curtis, Ward's method, cutree). They differ in the SECONDARY analysis that links cluster splits to environmental PCs (env PCA of 9 variables, PC1–PC9).

Survey (S17): clusters are already fixed, so the per-PC test is descriptive -- for each existing split within Division C it asks which environmental PC best separates the two sub-clusters. PERMANOVA was used for consistency with the survey's PERMANOVA/PERMDISP framework (S10, S17) and because Division C is large (97 samples).

Processing (P03): the tree walk is a decision rule -- it walks the dendrogram top-down and splits only where the two child groups differ environmentally, using a simple two-sample Wilcoxon rank-sum on PC scores, well suited to a binary split decision at the smaller processing sample size (78).

In short: survey = 'which PC characterizes this (fixed) split?' (PERMANOVA);
processing = 'is this split environmentally justified?' (Wilcoxon stopping rule).
Both correct across PCs with Benjamini–Hochberg. (The divergence also partly reflects that the two pipelines were developed separately; they can be harmonised on request.)

9. Cluster-split environmental testing – method comparison

Aspect	Survey	Processing
Primary clusters	Bray/Ward, cutree k=10 -> 11	Bray/Ward, cutree k=4
Env-node analysis	per-PC test at Division-C dendrogram nodes (S17)	top-down tree walk (P03)
Per-PC test	pairwise PERMANOVA (adonis2) on 1-D PC distance	Wilcoxon rank-sum on PC scores
Multiple-test correction	BH across PC1-PC9 within each node	BH across retained PCs at each node
Purpose	characterize which env axis separates a fixed split	decide whether to split (stopping rule)
Outcome	identifies the driving PC for each C sub-split	no env-significant split at root -> 1 cluster

9. Cluster-split environmental testing – survey vs processing

Both pipelines define primary clusters identically (Bray–Curtis, Ward, cutree). They differ in how each cluster split was tested against the environmental ordination.

Survey (S17): clusters are fixed; each split within Division C is tested per-PC by pairwise PERMANOVA + PERMDISP (all 9 PCs, BH per node). Descriptive – identifies which environmental axis drives an already-defined split.

Processing (P03): a top-down tree walk with a stopping rule; per-PC Wilcoxon rank-sum + BH (PCs up to 90% variance); a node is split only if ≥ 1 PC is significant. A decision rule – asks whether a split is environmentally justified.

Why different: survey Division C is large (97 samples) and embedded in a PERMANOVA/PERMDISP framework, so a multivariate permutation test was natural; processing is smaller (78) and needed a binary split rule, for which a simple two-sample Wilcoxon is well suited. Notably, the processing tree walk retained a SINGLE group (no env-significant split at the root), whereas survey Division C showed environmentally structured sub-splits. (The divergence also partly reflects independent development; the two can be harmonised.)

9a. Methods text – survey (Division C)

For the survey dataset, the finer structure within Division C was interpreted against an environmental ordination. The nine environmental variables (temperature, salinity, DO, NO₃, PO₄, silicate, NH₄, fluorescence and depth) were standardized and ordinated by PCA, and all nine components (PC1–PC9) were retained and interpreted from their loadings (e.g., PC1: nutrients, salinity and depth; PC2: DO versus temperature). The Division–C samples were re-clustered (Ward's method), and at each split of the resulting dendrogram (C1 vs C2, C1a vs C1b, C1b1 vs C1b2, C2a vs C2b, and C2b1 vs C2b2) the two descendant sub-clusters were compared along each PC separately: for every PC a one-dimensional Euclidean distance was computed from the sample PC scores and tested by pairwise PERMANOVA (adonis2, 999 permutations), with PERMDISP (betadisper) used to check homogeneity of dispersion. Within each node, the p-values across the nine PCs were adjusted with the Benjamini–Hochberg procedure, and the principal component(s) showing the most significant separation were regarded as the environmental axis driving that split.

9b. Methods text – processing

For the processing dataset, the community dendrogram was cut at $k = 4$ and the four clusters were grouped into two size-fraction divisions: free-living (Cluster 1) and particle-associated (Clusters 2–4). The environmental separability of the processing dendrogram was assessed separately using a top-down tree walk: at each node the two child groups (each ≥ 5 samples) were compared on the environmental principal components (those explaining $\geq 90\%$ of the cumulative variance) using Wilcoxon rank-sum tests with Benjamini–Hochberg correction, and a node was split only where at least one PC differed significantly (adjusted $p < 0.05$). This procedure retained a single group, indicating that — unlike the environmentally structured subdivisions within survey Division C — the processing clusters were not separated by a significant environmental gradient at the top level.

9c. Cluster-split env testing – method comparison

Aspect	Survey	Processing
Primary clusters	Bray/Ward, cutree k=10 -> 11	Bray/Ward, cutree k=4 (FL/PA)
Env test	per-PC test at fixed Division-C splits	top-down tree walk (recursive)
PCs used	all 9 (PC1-PC9)	>=90% cumulative-variance PCs
Per-PC test	pairwise PERMANOVA (adonis2) + PERMDISP	Wilcoxon rank-sum
Correction	BH across PC1-PC9 per node	BH across retained PCs per node
Split rule	none (all splits tested; descriptive)	split if >=1 PC significant (p<0.05)
Purpose	which env axis drives a fixed split	is a split env-justified?
Outcome	driving PC identified per C sub-split	single group (no env split at root)